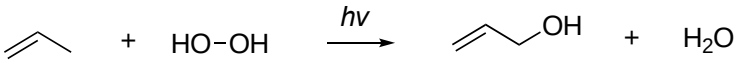
Organic Cume
November 11, 2022

1. Indicate which factors affect the rate of a reaction.
1). ΔG^o 2) ΔH^o 3). *E*a 4). temperature 5). concentration 6). K_{eq} 7). *k* 8). catalysts 9). ΔS^o

2. Given the bond dissociation energies shown, calculate the overall ΔH° for the following reaction.



Bond-Dissociation Energies for Homolytic Cleavage (kcal/mol)

H—H	104	°1 C—H	98	°1 C—Br	68
F—F	38	°2 C—H	95	°2 C—Br	68
Cl—Cl	58	°3 C—H	91	°3 C—Br	66
Br—Br	46	allylic C—H	87	°1 C—I	53
I—I	36	benzylic C—H	85	°2 C—I	53
HO—OH	51	°1 C—F	107	°3 C—I	50
H—F	136	°2 C—F	106	°1 C—OH	91
H—Cl	103	°3 C—F	106	°2 C—OH	91
H—Br	88	°1 C—Cl	81	°3 C—OH	91
H—I	71	°2 C—Cl	80	allylic C—OH	91
H—OH	119	°3 C—Cl	79	benzylic C—OH	91

3 Draw a reaction coordinate diagram for a reaction with the following criteria: a) exothermic, 3-step reaction; b) the first step is the rate-determining step; 2) the second intermediate is more stable than the first intermediate; d) the third step of the reaction is faster than the reverse of the second step.

4. A reaction is described overall as:

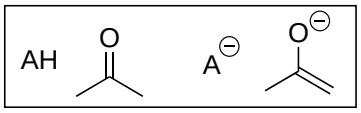
$$A + 2B \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2C$$

The following table shows initial rates of formation of C, labeled r_o, for various concentrations (in mM units, where mM = 10^{-3} M):

[A]	[B]	r_o (mM.s^{-1})
35	23	0.5 x 10^{-3}
70	46	2.0 x 10^{-3}
70	92	4.0 x 10^{-3}

1) Determine the empirical rate law in terms of formation of product C from this data, and clearly state your logic.
2)State the overall order and the order with respect to each component.
3) Calculate the rate constant (**be careful of units and coefficients**).

5. The condensation of acetone, $(CH_3)_2CO$, **AH**, in aqueous solution is base catalyzed. The base **B** reacts reversibly with **AH** to form the carbanion, $\mathbf{A^-}$. Then $\mathbf{A^-}$ reacts with AH to give the product, **P**. Determine a rate law for the overall reaction using steady state approximation, express the rate constant in terms of k_1, k_{-1}, k_2. **Hint: don't confuse catalysts with intermediates.**

$$AH + B \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} A^{\ominus} + \overset{\oplus}{B}H$$

$$AH + A^{\ominus} \xrightarrow{k_2} P$$